

Chapter 2

E-Commerce: Mechanisms, Platforms, and Tools

Learning Objectives

1. Describe the major electronic commerce (EC) activities and processes and the mechanisms that support them.
2. Define e-marketplaces and list their components.
3. List the major types of e-marketplaces and describe their features.
4. Describe electronic catalogs, search engines, and shopping carts.
5. Describe the major types of auctions and list their characteristics.

Learning Objectives

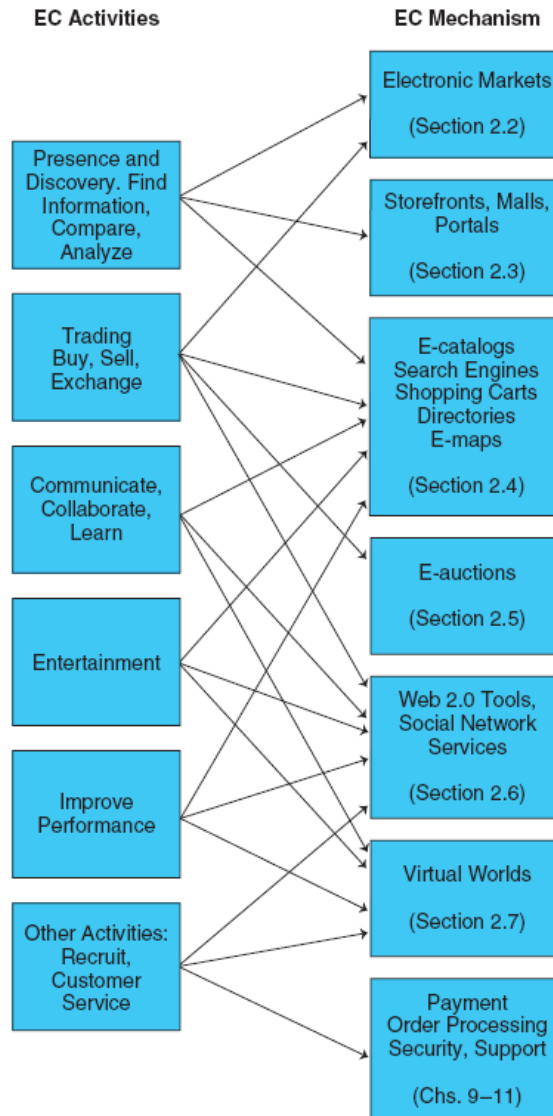
6. Discuss the benefits and limitations of e-auctions.
7. Describe bartering and negotiating online.
8. Describe virtual communities.
9. List the major Web 2.0 tools and their use in EC.
10. Describe social networks as an EC mechanism.
11. Understand virtual worlds and their use in EC.
12. Describe Web 3.0 and define Web 4.0.

Electronic Commerce Mechanisms: An Overview

- **EC ACTIVITIES AND SUPPORT MECHANISMS**
- **The EC trading activities are divided into six categories**
- **Each activity is supported by one or more EC mechanisms**

EXHIBIT 2.1

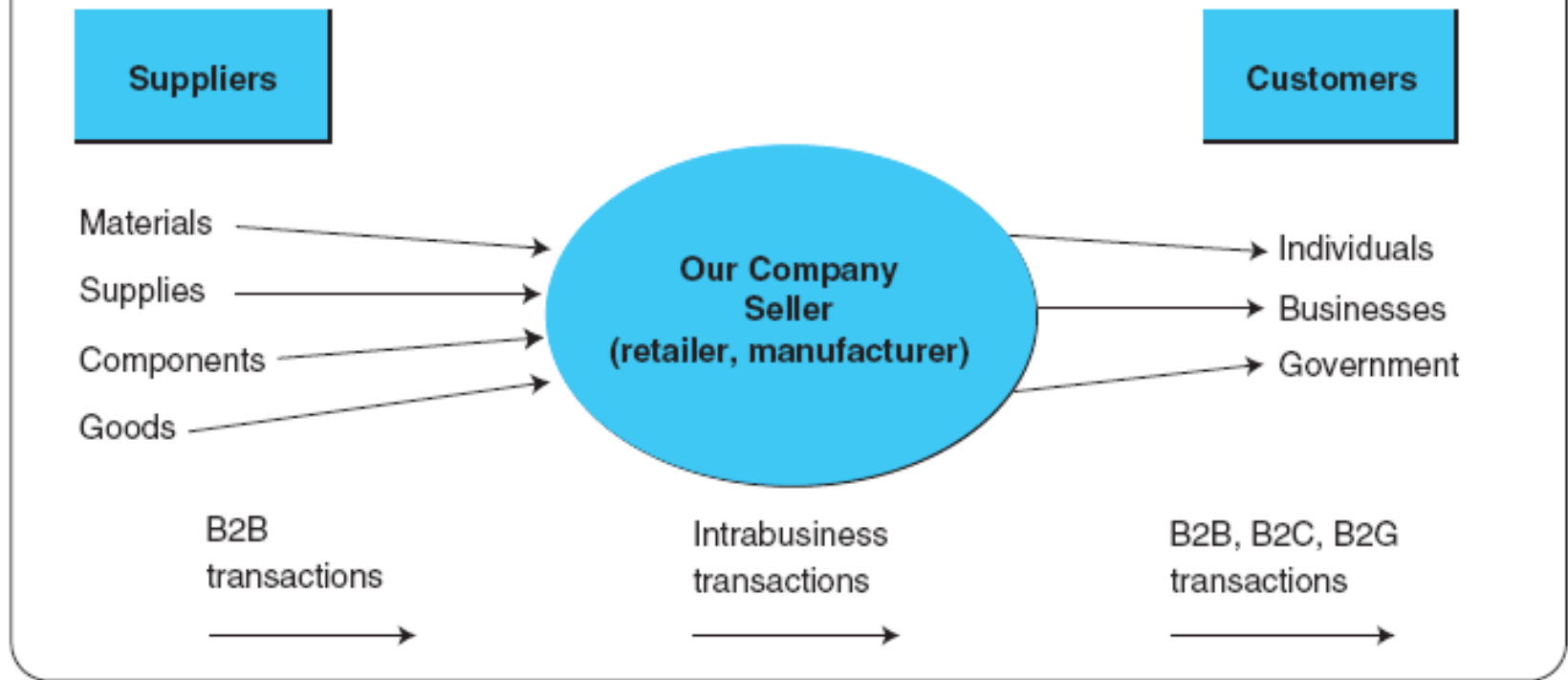
The EC Activities–Mechanism Connection



Electronic Commerce Mechanisms: An Overview

- **SELLERS, BUYERS, AND TRANSACTIONS**
- **a seller(retailer,wholesaler or manufacturer) sells to customers.**
- **The seller buys from suppliers either raw materials (as a manufacturer) or components for assembly (as an assembler) ,or finished goods (as retailer).**

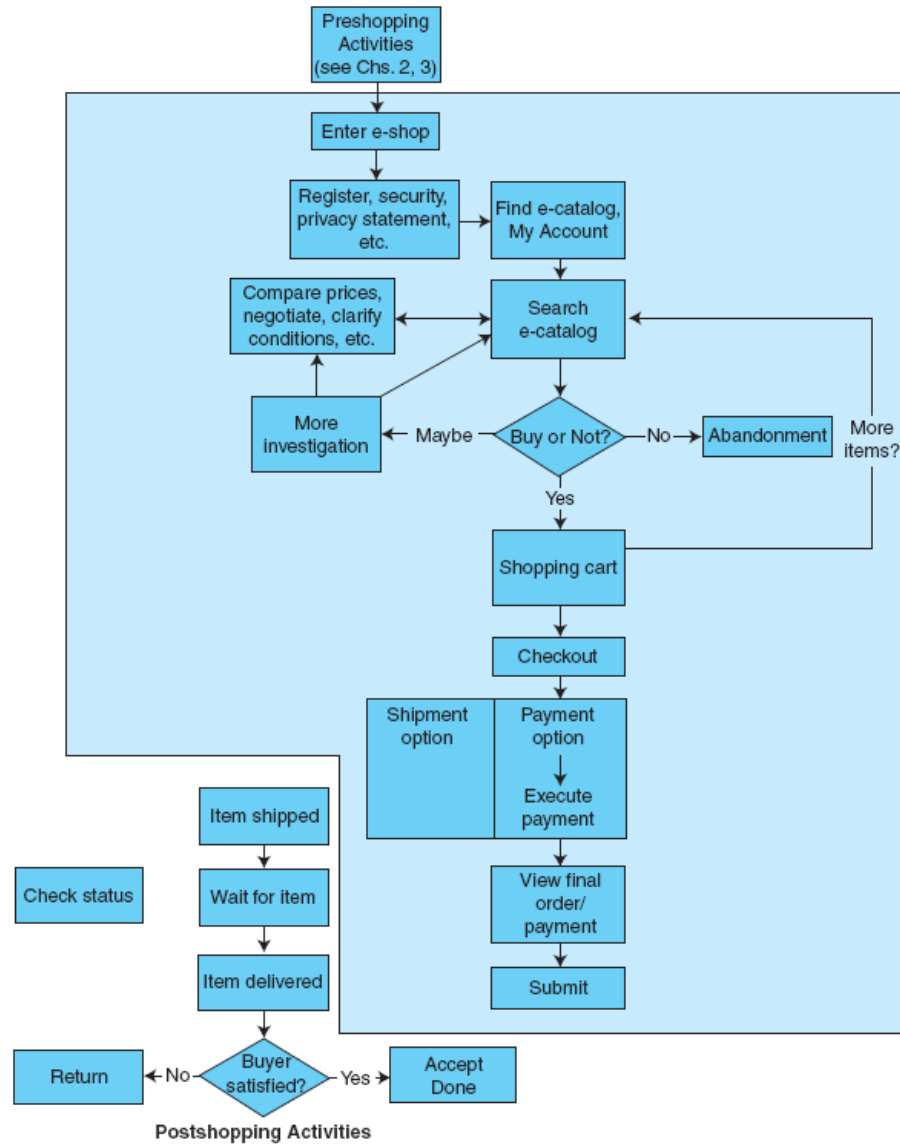
EXHIBIT 2.2 EC Participants and Components



Electronic Commerce Mechanisms: An Overview

- **The Purchasing Process**
- **Customers buy goods online in different ways.**
- **The most common way is purchasing from catalogs at fixed prices.**
- **Some times prices may be negotiated or discounted.**
- **Another way to determine price is dynamic pricing, which refers to nonfixed prices such as those in stock markets.**

EXHIBIT 2.3 Buying Process in an E-Market



E-Marketplaces

- **e-marketplace**

An online market in which buyers and sellers exchange goods or services;

- the three types of e-marketplaces are private, public, and consortia

- **marketplace**

A marketplace in which sellers and buyers exchange goods and services for money (or for other goods and services), but do so electronically

E-Marketplaces

- The major components and players in a marketplaces are:

- **Customers**
- **Sellers**
- **Products and services**
 - **digital products**

Goods that can be transformed to digital format and delivered over the Internet

- **Infrastructure**

electronic networks, databases, hardware, software, and more.

E-Marketplaces

- **front end**

The portion of an e-seller's business processes through which customers interact, including the seller's portal, electronic catalogs, a shopping cart, a search engine, and a payment gateway

- **back end**

The activities that support online order fulfillment, inventory management, purchasing from suppliers, payment processing, packaging, and delivery

- **intermediary**

A third party that operates between sellers and buyers

- **Support services:** such as security, Payment, certification, etc.

E-Marketplaces

- **TYPES OF E-MARKETPLACES:**

1. **Private E-Marketplaces:** they are owned and operated by a single company

- **sell-side e-marketplace**

A private e-marketplace in which one company sells either standard and/or customized products to individuals or companies. It is considered to be *one-to-many*

- **buy-side e-marketplace**

A private e-marketplace in which one company makes purchases from many suppliers . It is considered to be *many-to-one*

E-Marketplaces

2. **Public E-Marketplaces:**

- They are owned by a third party (not a seller or a buyer) or by a group of selling and buying companies (a consortium)
- They serve many sellers and many buyers
- They are known as exchanges

Customer Shopping Mechanisms: Webstores, Malls, and Portals

- **Webstore (storefront)**

A single company's website where products or services are sold;

- Several mechanisms are included in a Webfront:

- Electronic catalog
- Search engine
- Electronic cart
- E-auction facilities
- Payment gateway
- Shipment court
- Customer services

Customer Shopping Mechanisms: Storefronts, Malls, and Portals

- **e-mall (online mall)**

An online shopping center where many online stores are located

- **TYPES OF STORES AND MALLS**

- General stores/malls
- Specialized stores/malls
- Regional versus global stores
- Pure-play versus click-and-mortar stores

Customer Shopping Mechanisms: Storefronts, Malls, and Portals

- **Web (information) portal**

A single point of access, through a Web browser, to critical business information located inside and outside (via Internet) an organization

- **Types of Portals**

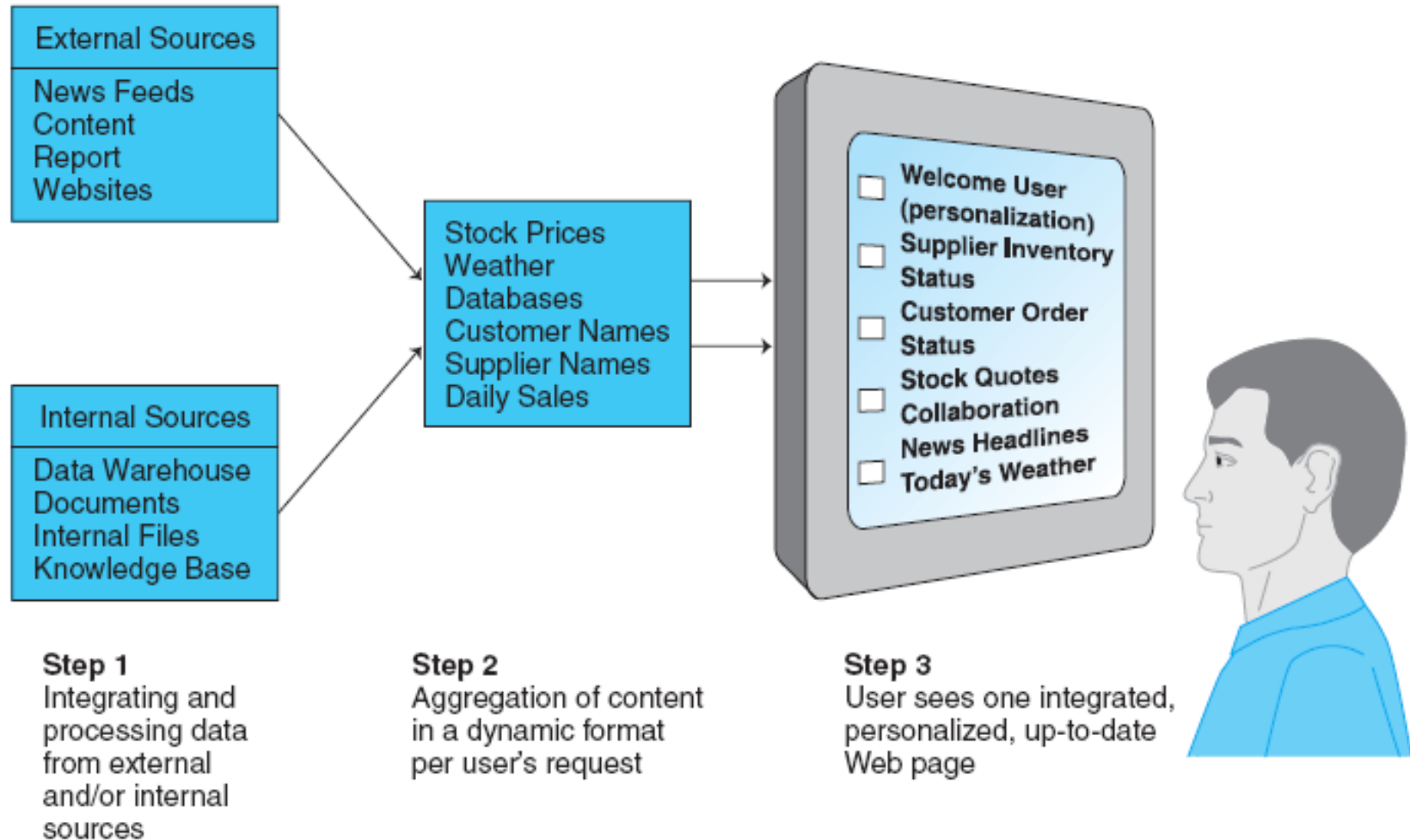
- Commercial (public) portals
- Corporate portals
- Publishing portals
- Personal portals
- **mobile portal**

A portal accessible via a mobile device.

- **voice portal**

A portal accessed by telephone or cell phone.

EXHIBIT 2.5 How a Portal Works



Customer Shopping Mechanisms: Storefronts, Malls, and Portals

- **THE ROLES AND VALUE OF INTERMEDIARIES IN E-MARKETPLACES**
 - Intermediaries, such as brokers, play an important role in commerce by providing value-added activities and services to buyers and sellers.
 - The two major types of online intermediaries are brokers and infomediaries.

Customer Shopping Mechanisms: Storefronts, Malls, and Portals

- **Brokers**

- A broker in EC is a person or a company that facilitates transactions between buyers and sellers.
- The following are different types of brokers:
 - Trading
 - Organization of online malls
 - Comparison agent.
 - Shopping aids provider
 - Matching services

Customer Shopping Mechanisms: Storefronts, Malls, and Portals

- **infomediaries**

Electronic intermediaries that provide and/or control information flow in cyberspace, often aggregating information and selling it to others

- **Distributors in B2B**

A special type of intermediary in e-commerce is the B2B e-distributor

- **e-distributor**

An e-commerce intermediary that connects manufacturers with business buyers (customers) by aggregating the catalogs of many manufacturers in one place—the intermediary's website

Merchant Solutions: Electronic Catalogs, Search Engines, and Shopping Carts

- **electronic catalogs (e-catalogs)**

The presentation of product information in an electronic form; the backbone of most e-selling sites

- **Online Catalogs Versus Paper Catalogs**

Merchant Solutions: Electronic Catalogs, Search Engines, and Shopping Carts

- **EC SEARCH ACTIVITIES, TYPES, AND ENGINES**

- **Types of EC Searches**

- **Internet/Web Search**
- **enterprise search**

The practice of identifying and enabling specific content across the enterprise to be indexed, searched, and displayed to authorized users

Merchant Solutions: Electronic Catalogs, Search Engines, and Shopping Carts

- **desktop search**

Search tools that search the contents of a user's or organization's computer files, rather than searching the Internet

The emphasis is on finding all the information that is available on the user's PC, including Web browser histories, e-mail archives, and word-processed documents, as well as in all internal files and databases.

Merchant Solutions: Electronic Catalogs, Search Engines, and Shopping Carts

- **search engine**

A computer program that can access databases of Internet resources, search for specific information or key words, and report the results

- **Software (Intelligent) Agents**
- **Questions and Answers Online**
- **Voice-Powered Search**
- **Visual Shopping Search Engine**
- **Video and Mobile Search**
- **Social Network Search Engines**

Merchant Solutions: Electronic Catalogs, Search Engines, and Shopping Carts

- **electronic shopping cart**

An order-processing technology that allows customers to accumulate items they wish to buy while they continue to shop

Auctions, Bartering, and Negotiating Online

- **auction**

A competitive process in which a seller solicits consecutive bids from buyers (forward auctions) or a buyer solicits bids from sellers (backward auctions); prices are determined dynamically by the bids

- **dynamic pricing**

Prices that change based on supply and demand relationships at any given time

Auctions, Bartering, and Negotiating Online

- **TRADITIONAL AUCTIONS VERSUS E-AUCTIONS**

- **Limitations of Traditional Offline Auctions**
- **electronic auctions (e-auctions)**

Auctions conducted online

Auctions, Bartering, and Negotiating Online

- **TYPES OF AUCTIONS**

- **One Buyer, One Seller**
- **One Seller, Many Potential Buyers**
 - **forward auction**

An auction in which a seller entertains bids from buyers; bidders increase price sequentially

Auctions, Bartering, and Negotiating Online

- **One Buyer, Many Potential Sellers**

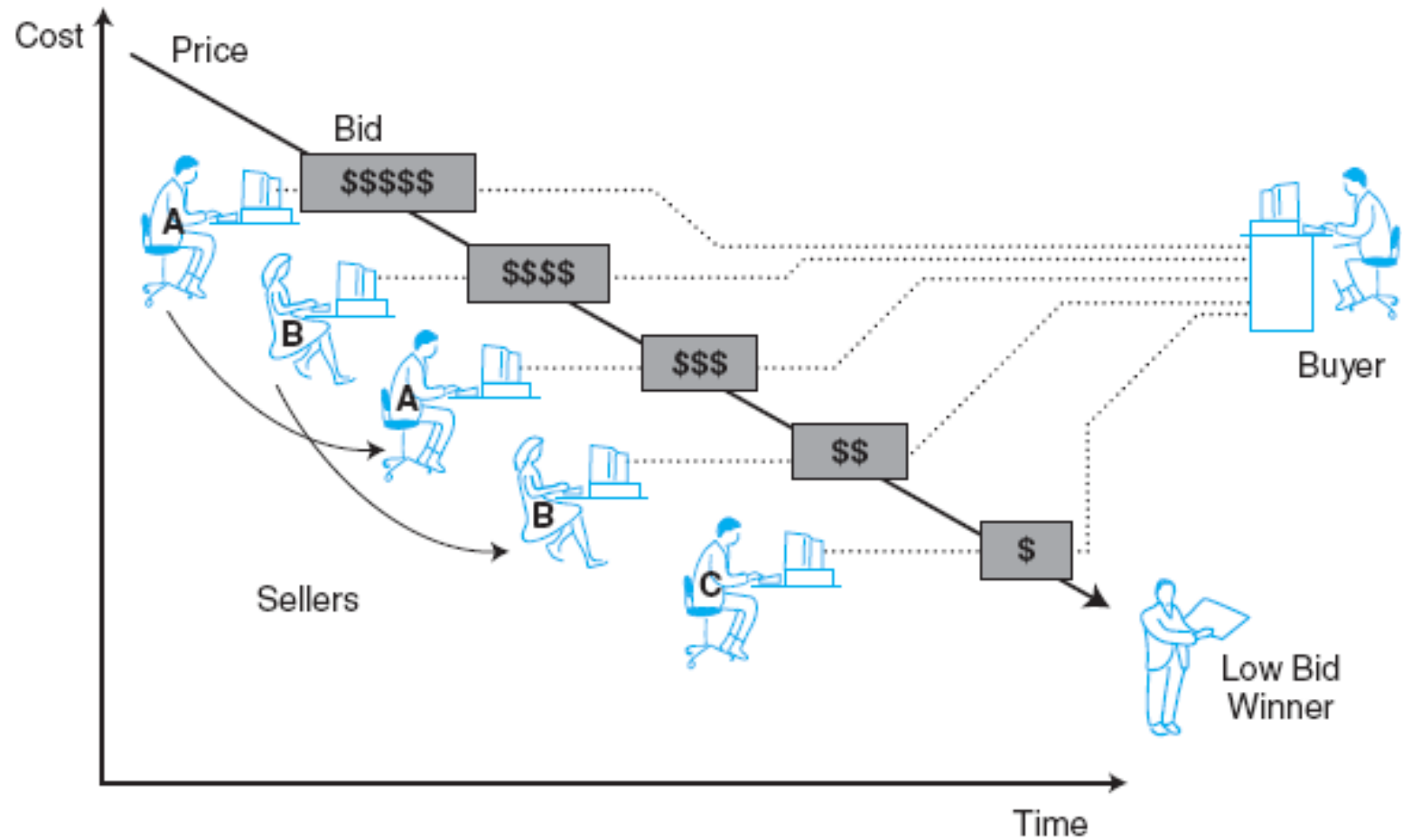
- **reverse auction (bidding or tendering system)**

Auction in which the buyer places an item for bid (tender) on a request for quote (RFQ) system, potential suppliers bid on the job, with the price reducing sequentially, and the lowest bid wins; primarily a B2B or G2B mechanism

- **name-your-own-price model**

Auction model in which a would-be buyer specifies the price (and other terms) he or she is willing to pay to any willing and able seller; a C2B model that was pioneered by Priceline.com.

EXHIBIT 2.6 The Reverse Auction Process



Auctions, Bartering, and Negotiating Online

- **Many Sellers, Many Buyers**

- **double auction**

An auction in which multiple buyers and their bidding prices are matched with multiple sellers and their asking prices, considering the quantities on both sides

- **penny auction**

A formal auction in which participants pay a nonrefundable small fee for each bid; bid level changes by small increments

EXHIBIT 2.7 Benefits of E-Auctions

Benefits to Sellers

- Increased revenues from broadening bidder base and shortening cycle time. Can sell anywhere globally.
- Opportunity to bargain instead of selling at a fixed price. Can sell any time and conduct frequent auctions.
- Optimal price setting determined by the market (more buyers, more information).
- Sellers can gain more customer dollars by offering items directly (saves on the commission to intermediaries; also, physical auctions are very expensive compared with e-auctions).
- Can liquidate large quantities quickly.
- Improved customer relationship and loyalty (in the case of specialized B2B auction sites and electronic exchanges).

Benefits to Buyers

- Opportunities to find unique items and collectibles.
- Entertainment. Participation in e-auctions can be entertaining and exciting.
- Convenience. Buyers can bid from anywhere, even with a cell phone; they do not have to travel to an auction place.
- Anonymity. With the help of a third party, buyers can remain anonymous.
- Possibility of finding bargains, for both individuals and organizations.

Benefits to E-Auctioneers

- Higher repeat purchases. Jupiter Research (jupiterresearch.com) found that auction sites, such as eBay, tend to garner higher repeat-purchase rates than the top B2C sites, such as Amazon.com.
- High “stickiness” to the website (the tendency of customers to stay at sites longer and come back more often). Auction sites are frequently “stickier” than fixed-priced sites. Stickier sites generate more ad revenue for the e-auctioneer.
- Easy expansion of the auction business.

Auctions, Bartering, and Negotiating Online

- **Limitations of E-Auctions**

- Minimal Security
- Possibility of Fraud
- Limited Participation

Auctions, Bartering, and Negotiating Online

● IMPACTS OF AUCTIONS

- The following are some representative impacts of e-auctions.
 - Auctions as a Social Mechanism to Determine a Price
 - Auctions as a Highly Visible Distribution Mechanism
 - Auctions as an EC Component in a Business Model
 - Auctions for Profit for Individuals

Auctions, Bartering, and Negotiating Online

- **ONLINE BARTERING**

- **bartering**

- The exchange of goods and services

- **e-bartering (electronic bartering)**

- Bartering conducted online, usually in a bartering exchange

- **bartering exchange**

- A marketplace in which an intermediary arranges barter transactions

- **ONLINE NEGOTIATING**

Social Software Tools:

From Blogs to Wikis to Twitter

- **social software**

A software product that enables people to rendezvous, engage connect, and collaborate through computer-mediated communication e.g facebook, Flickr, youtube, amazon.com, eBay

- **blog**

A personal website that is open to the public to read and to interact with; dedicated to specific topics or

issues in which owner expresses his or her feeling or opinions. Blogs can result in two way communication and collaboration, group discussions and so on.

- **vlog (or video blog)**

A blog with video content

Social Software Tools: From Blogs to Wikis to Twitter

- **Building Effective Blogs:**
 - It is becoming easier to build blogs.
 - Programs from blogger.com are very user –friendly.
 - Blog space is free.
 - Bloggers(the people who create and maintain blogs) are handed a fresh space on their website to write in each day.
 - They can easily edit, add ,entries and broadcast whatever they want by simply clicking “send”.
- **Commercial Uses of Blogs**
 - The blog concept has transferred to the commercial world
- **Potential Risks of Corporate Blogs**
 - some people see risk in commercial blogging.

Social Software Tools: From Blogs to Wikis to Twitter

- **microblogging**

A form of blogging that allows users to write messages (usually up to 140 characters) and publish them, either to be viewed by anyone or by a restricted group that can be chosen by the user; these messages can be submitted by a variety of means, including text messaging, instant messaging, e-mail, MP3, or just on the Web

- **Twitter**

A free microblogging service that allows its users to send and read other users' updates

Social Software Tools: From Blogs to Wikis to Twitter

- **tweets**

Text-based posts up to 140 characters in length posted to Twitter

- **The Essentials of Twitter for Business**

- The Major Benefits of Twitter
- Examples of Twitter as Enterprise Tools

Social Software Tools: From Blogs to Wikis to Twitter

- **wiki (wikilog)**

A blog that allows everyone to participate as a peer; anyone may add, delete, or change content

- **Business Applications of Wikis**

Social Software Tools:

From Blogs to Wikis to Twitter

- **MECHANISM AIDS FOR WEB 2.0 TOOLS: TAGS, FOLKSONOMY, MASHUPS, AND SOCIAL BOOKMARKS**

- **tag**

A nonhierarchical key word or term assigned to a piece of information (such as an Internet bookmark, digital image, video clip, or any computer document)

- **folksonomy (collaborative tagging, social tagging)**

The practice and method of collaboratively creating, classifying, and managing tags to annotate and categorize content

Social Software Tools: From Blogs to Wikis to Twitter

- **mashup**

Combination of two or more websites into a single website that provides the content of both sites (whole or partial) to deliver a novel product to consumers

- **social bookmarking**

Web service for sharing Internet bookmarks; the sites are a popular way to store, classify, share, and search links through the practice of folksonomy techniques on the Internet and intranets

Virtual Communities and Social Networks

- **virtual community**

A group of people with similar interests who interact with one another using the Internet

- The interaction takes place over a computer network, mainly the internet.
- virtual communities parallel typical physical communities, such as neighborhood, clubs, or associations, but people do not meet face to face.
- Instead, they meet online.
- It is a social network organized around a common interest, idea, task, or goal, members interact across time, geographic, and organizational boundaries to develop personal relationships.

Virtual Communities and Social Networks

- virtual communities offer several ways for members to interact, collaborate, and trade.
- Most virtual communities are internet-based known as Internet communities.
- Hundreds of thousands of communities exist on the internet, and the number is growing rapidly.
- **Types of virtual Communities**

Table 2.3 Types of virtual communities

Community type	Description
Transaction and other business activities	Facilitate buying and selling Combines an information portal with an infrastructure for trading. Members are buyers, sellers, intermediaries, etc., who are focused on a specific commercial area (e.g., fishing)
Purpose or interest	No trading, just exchange of information on a topic of mutual interest. Examples: Investors consult The Motley Fool (fool.com) for financial advice; music lovers go to mp3.com
Relations or practices	Members are organized around certain life experiences. Examples: ivillage.com caters to women and seniornet.com is for senior citizens. Professional communities also belong to this category. Examples: aboutus.org/ Isworld.org is a space for information systems faculty, students, and professionals
Fantasy/role playing	Members share imaginary environments. Examples: sports fantasy teams at espn.com ; see games.yahoo.com , horseracegame.com
Social networks	Members communicate, collaborate, create, share, form groups, entertain, and more. Facebook is the leader
Virtual worlds	Members use avatars to represent themselves in a simulated 3-D environment where they can play games, conduct business, socialize, and fantasize about whatever they like

Virtual Communities and Social Networks

- **Other Classifications of Virtual Communities**
 - **Public Versus Private Communities**
- **Public communities**
 - their membership is open to anyone.
 - The owner of the community may be a privately held corporation or a public one.
 - .e.g facebook belong to public category
- **Private communities**
 - belong each to a company, an association or a group of companies and their membership is limited to people who meet certain requirement.
 - e.g (work for a particular employer or work in a particular profession).
 - Private communities may be internal (e.g only employees can be members) or external(for customers)

Virtual Communities and Social Networks

- **Classification Categories**
 - **Another option is to classify the members as traders, players, just friends, or friends in need.**

Virtual Communities and Social Networks

● ONLINE SOCIAL NETWORKS

- **Social networks** is a place where people create their own space, or homepage, on which they write blog; post picture, videos, music, share ideas, link other web locations, can tag the content they create and post it with key words.
- **The Size of Social Network Sites**
- Social network sites are growing rapidly, with some having over 100 million members. e.g. facebook had 750 million members in June 2011.
- **A Global Phenomenon**
- Although facebook and MySpace attract the majority of media attention in the United States, they do have many members in other countries. Other social network sites are proliferating and growing in popularity worldwide.

Virtual Communities and Social Networks

- **Representative Capabilities and Services Provided by Social Network Sites**
- **Social network sites provide many capabilities and services such as**
 - **create a circle of friends,**
 - **discussion forums,**
 - **share photo,video,documents,etc,**
 - **construct a Web page**
 - **Messages**

Virtual Communities and Social Networks

- **business-oriented social networks**

A social network whose major interest is business topics and whose members are professional people; such networks are used mostly for creating contacts, providing requirements, and enlisting members' support for problem solving and knowledge sharing

- Examples: linkedIn

Virtual Communities and Social Networks

- **Example of a Business-Oriented Social Network**
- User can visit the business-oriented social network websites to
 - establish new business contacts,
 - expand and manage their contact's networks,
 - identify experts and receive advice,
 - organize meetings and events.
- **Some Capabilities of Business-Oriented Networks**
- By using these websites company can
 - encourage consumers to rate and comment of products,
 - allow consumers to create their own topic areas and build communities and shared interest related to a company's products.
- **Business Models and Services Related to Social Networking**
- Social networking sites provide innovative business models, ranging from customer reviews of food and night life in India to users who dress up dolls that look like celebrities.

Virtual Communities and Social Networks

- **Social Network Analysis Software**

- These software is used to identify, represent, analyze, visualize network nodes (e.g agents, organizations or knowledge)

- **Xanga**

- Xanga.com host blogs, photoblogs and social networking profiles.

- **Digg**

- Digg.com is a community based website with an emphasis on technology and science articles.

Virtual Communities and Social Networks

- **Mobile Social Commerce**

- It is growing faster than any other type of EC computing
- According to The Retail Bulletin (2012), 64% smartphone consumers used them to shop online

- **mobile social networking**

- It refers to social networking where members chat and connect with one another using any mobile device

Virtual Communities and Social Networks

- **Recent Innovative Tools and Platforms for Social Networking:**
 - Snapchat.com
 - Whatsapp.com
 - Instagram.com
 - Etc.

Virtual Communities and Social Networks

- **Mobile Community Activities**

- **Users can use their mobile devices to**
 - create their profiles,
 - make friends participate in chat rooms,
 - create chat rooms,
 - hold private conversations,
 - share photos, videos and blogs.

- **SOCIAL NETWORK SERVICES**

- Social network services are companies that host social communities.
- They are known in brief as social networks.
- social-oriented networks are Facebook and MySpace.
- LinkedIn is a business-oriented network.

Virtual Worlds as an Electronic Commerce Platforms

- **virtual world**

A user-defined world in which people can interact, play, and do business; the most publicized virtual world is Second Life

- Virtual worlds are not limited to games.
- Players can create a character to travel between buildings, towns, and even worlds to carry out business or leisure (free , holdidays) activities.

- **Avatars**

- Virtual worlds are populated with interactive animated characters—software agents known as Avatars.

Virtual Worlds as an Electronic Commerce Platforms

- Avatars are animated computer characters that exhibit humanlike movements and behaviors
- Advanced avatars can “speak” and display behaviors such as gestures and facial expressions.
- They can be fully automated to act like robots.
- Many companies use avatars as tour guides or to staff virtual reception desks.
- Avatars are aimed at making the human-computer interface more natural.

Virtual Worlds as an Electronic Commerce Mechanism

- **BUSINESS ACTIVITIES AND VALUE IN VIRTUAL WORLDS**
- There are several types of business activities in virtual worlds.
 - Creating and managing a virtual business
 - Conducting regular business activities
 - Providing services for those who build, manage, or make money with virtual properties.
- Examples of how companies use virtual worlds in businesses:
 - Market Research
 - guide and advise passengers in airports

Virtual Worlds as an Electronic Commerce Mechanism

● Virtual Shopping

- You go shopping with friends even while you remain miles apart.
- In reality, you are all at your computer terminal, but on the screen you would be transported to a digital replica of the shopping center
- As you walk at a virtual jeans store, web cameras in the real store let you see how crowded in actually is, your avatar ,set to your body's measurements, tries on the jeans. you might buy things online or visit the physical store later.

● Trading Virtual Properties

- Trading virtual properties is a very popular activity in second life.
- haboo.com sells more virtual furniture worldwide
- the furniture is designed by teens on the site who decorate their rooms with furniture.
- The kids are buying not only furniture but also clothes, bags etc.

EMERGING EC PLATFORMS:

AUGMENTED REALITY AND CROWDSOURCING

● **Augmented Reality**

- Augmented reality is “a live, copy, view of a physical, real-world environment whose elements are augmented(or supplemented) by computer-generated sensory input such as sound, video, graphics or GPS data”.
- The computerized layer can be seen through an application on mobile devices such as smartphones, webcams, or 3D glasses (including 3D TV)

● **Crowdsourcing**

- Another platform for e-commerce is crowdsourcing.
- The term crowd refers to any group of people such as a group of consumers, employees of a corporation, or members of a social network who offer expertise.
- A Crowd is frequently referred to a large group.
- Crowdsourcing utilizes crowds to collectively execute tasks such as solving problems, innovating, or getting large projects completed by dividing the work among many people.

● ***Benefits of Crowdsourcing***

EMERGING EC PLATFORMS: AUGMENTED REALITY AND CROWDSOURCING

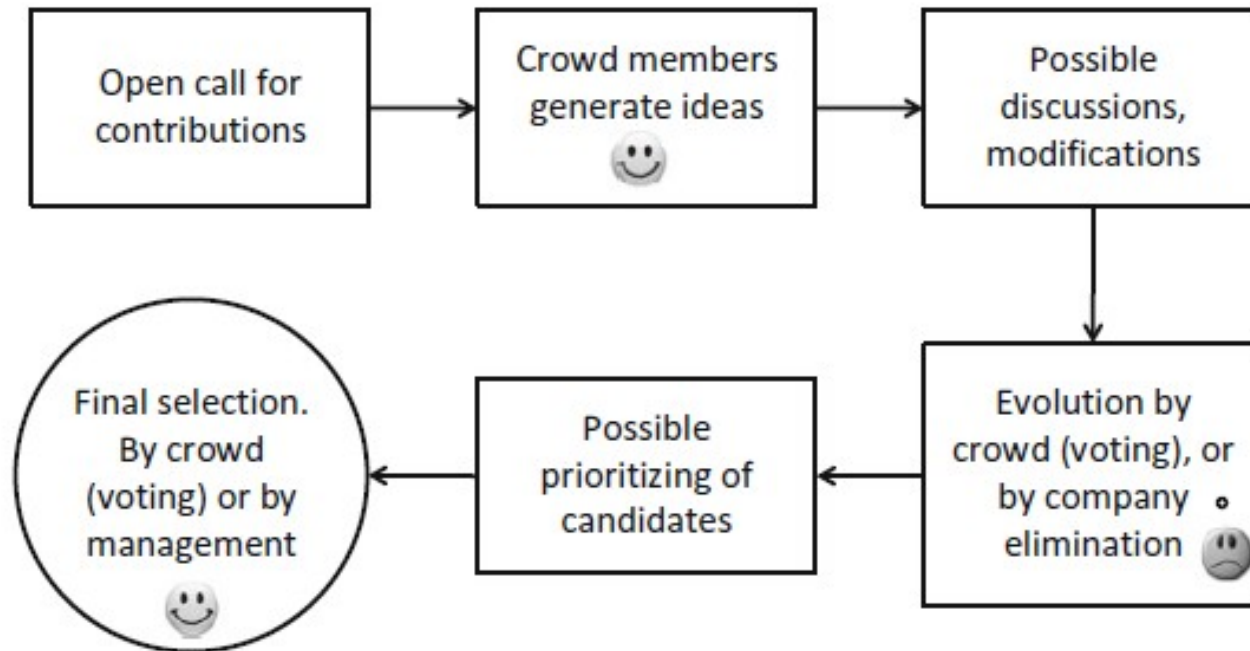


Figure 2.6 A typical crowdsourcing process

The Future: Web 3.0 and Web 4.0

- **Web 3.0**

- A term used to describe the future of the World Wide Web;
- is projected to deliver a new generation of business applications that will see business and social computing converge
- Web 3.0 could change the manner in which people live and work as well as the organizations where they work, and it may even revolutionize social networking

- **Web 3.0 and Semantic Web**

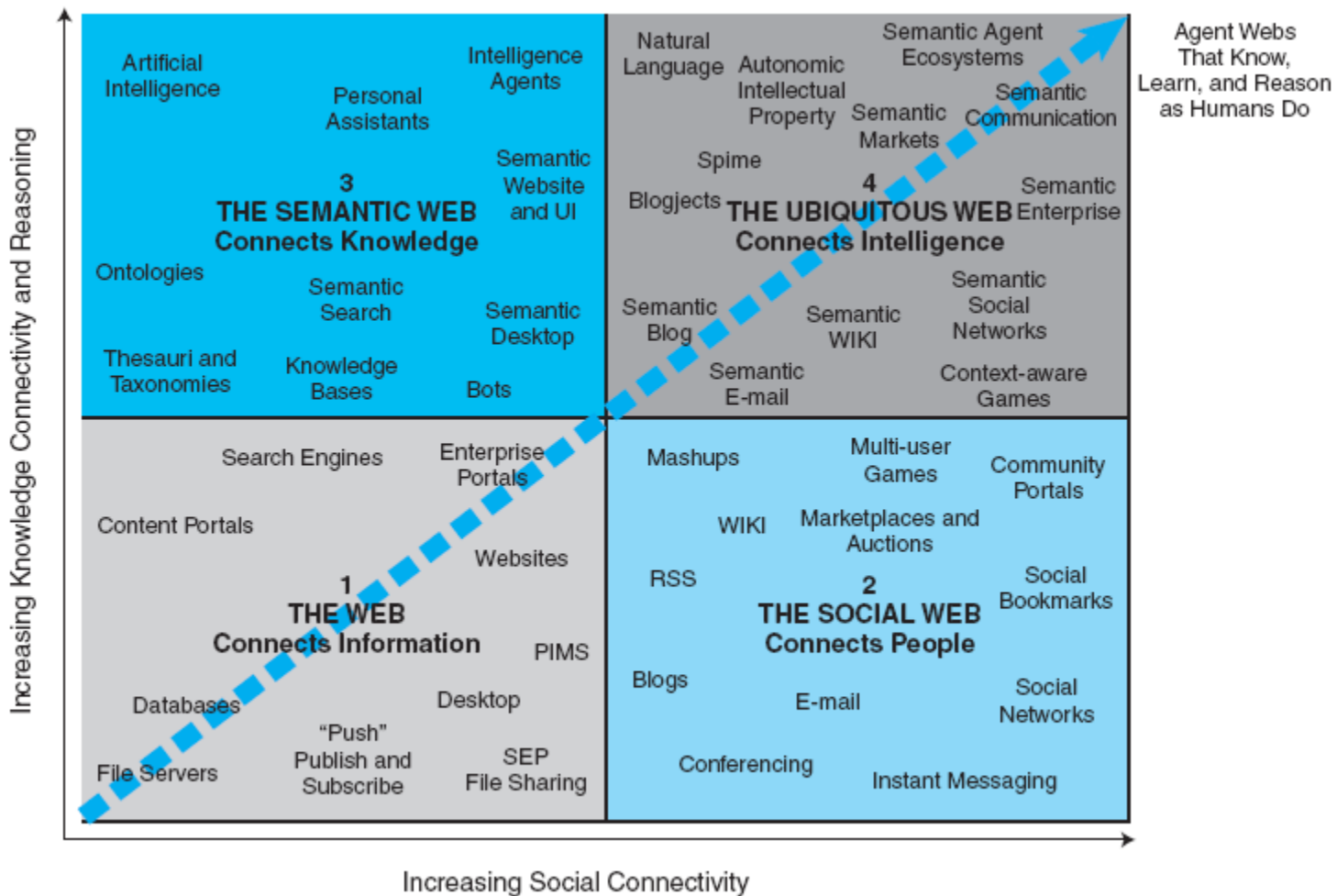
An evolving extension of the Web in which Web content can be expressed not only in natural language, but also in a form that can be understood, interpreted, and used by intelligent computer software agents, permitting them to find, share, and integrate information more easily

The Future: Web 3.0 and Web 4.0

- **Web 4.0**

The Web generation after Web 3.0 that is still an unknown entity; however, it is envisioned as being based on islands of intelligence and as being ubiquitous

EXHIBIT 2.10 Evolution from Web 1.0 to Web 4.0



Managerial Issues

1. Should we use auctions for selling?
2. Should we barter?
3. How do we select merchant software?
4. How can we use Facebook and other social networks in our business?
5. How shall we start using Web 2.0 tools?
6. Shall we take part in virtual worlds?
7. How should we deal with Web 2.0 risks?

Summary

1. Activities and mechanisms
2. E-marketplaces and their components
3. The major types of e-marketplaces
4. Electronic catalogs, search engines, and shopping carts
5. Types of auctions and their characteristics
6. The benefits and limitations of auctions

Summary

7. Bartering and negotiating
8. The structure and role of virtual communities
9. Web 2.0 tools
10. Social networks as an EC mechanism
11. Virtual worlds
12. Web 3.0 and Web 4.0